

JOSE ANGEL SANCHEZ GOMEZ

Assistant Professor, Department of Statistics

(+1)919-697-6202 ◇ josesa@ucr.edu

PROFESSIONAL EXPERIENCE

Assistant Professor <i>Department of Statistics, University of California at Riverside (UCR)</i>	August 2023 - current <i>Riverside, CA</i>
Data Science for the Social Good Fellow <i>German Research Center for Artificial Intelligence (DFKI)</i>	June-August 2022 <i>Kaiserslautern, Germany</i>

EDUCATION

University of North Carolina at Chapel Hill, USA <i>Ph.D. in Statistics</i>	August 2018 - June 2023
The University of Guanajuato, Mexico <i>B.Sc. in Mathematics</i>	August 2013 - June 2018

RESEARCH MANUSCRIPTS

Peer-Reviewed Articles

- **Sánchez Gómez, J.**, Zhang, E., Liu, Y. (2025) *Effective Permutation Tests for Differences Across Multiple High-Dimensional Correlation Matrices*. Journal of Computational and Graphical Statistics, 1–10 <https://doi.org/10.1080/10618600.2025.2550527>.
- Cordero, J. M., Henn, T., Holtel, F., **Sánchez Gómez, J.**, Arenas, D., Sipka, A. Vollmer, S. (2025). *Developing Effective and Value-Aligned AI Tools for Journalists: 12 Critical Questions to Reflect upon*. Accepted in Journalism Practice <https://doi.org/10.1080/17512786.2025.2465894>.
- **Sánchez Gómez, J.**, Mo, W., Zhao, J., Liu, Y. (2025). *Hub Detection in Gaussian Graphical Models*. Journal of the American Statistical Association 120(552), 2397-2409. <https://doi.org/10.1080/01621459.2025.2453250>.
- Griffith, A., **Sánchez Gómez, J.**, Castillo, K. (2024). *Hurricane Irma Linked to Coral Skeletal Density Shifts on The Florida Keys Reef Tract*. Integrative And Comparative Biology. <https://doi.org/10.1093/icb/icae128>
- Chowdhury, S., Clause, N., Memoli, F., **Sánchez Gómez, J.**, Wellner, Z. (2020) *New Families of Stable Simplicial Filtration Functors*. Topology and its Applications, 279(1), 107254. <https://doi.org/10.1016/j.topol.2020.107254>

In Preparation and In Progress

- **Sánchez Gómez, J.**, O'Rourke, H., Brewer, G. (2026) *Estimating Moderator Effects on Multiple Subject Intensive Longitudinal Studies*. In preparation since October 2024.
- **Sánchez Gómez, J.**, Mo, W., Zhao, J., Liu, Y. (2026) *Joint Eigenspace Analysis: Perturbation Theory and Applications*. In progress since January, 2026.
- **Sánchez Gómez, J.**, Mo, W., Zhao, J., Liu, Y. (2026) *Identifying Common Hubs in Multiple Gaussian Graphical Models*. Under review at the 43rd International Conference on Machine Learning (ICML-2026).

- Cai, X., Gao, Q., **Sánchez Gómez, J.**, Au, K. (2026) *Cluster-assisted Gene Isoform Quantification: Theory and Practice*. In preparation since October 2023.
- Seongoh, P., **Sánchez Gómez, J.** (2026) *Transfer Learning for High-Dimensional Gaussian Graphical Models*. In preparation since September, 2024.
- **Sánchez Gómez, J.**, Whitney, A. (2026) *Machine Learning to Predict Local Government Deficits*. Under review at the Journal of Public Budgeting, Accounting & Financial Management.

TEACHING EXPERIENCE

At the University of California at Riverside

- STAT 293: High-Dimensional Statistics (Fall 2025, Fall 2024, Fall 2023)
- STAT 107: Introduction to Statistical Computing with R (Fall 2025, Winter 2025, Winter 2024)
- STAT 155: Probability and Statistics for Science and Engineering (Winter 2024, Spring 2025, Spring 2026)

SELECTED PRESENTATIONS

- Invited Speaker “Identifying Common Hubs in Multiple Gaussian Graphical Models” at the Joint Statistical Meeting of the American Statistical Association in Nashville, Tennessee, August, 2025.
- Invited Speaker “Hub Detection in Gaussian Graphical Models” at the Statistical Learning and Data Science meeting of the American Statistical Association (SLDS) in Newport Beach, CA, November, 2024.
- Invited Speaker “Hub Detection in Gaussian Graphical Models” at the Institute of Mathematical Statistics Asia-Pacific Rim Meeting (IMS-APRM) at University of Melbourne, January, 2024.
- Presentation “Searching for New Topological Filtrations, Data” in the *Joint Meeting of the Mexican and Colombian Mathematical Societies* hosted in Barranquilla, Colombia, June, 2018.
- Poster “A First Approach to a General Theory of Filtration Functors” at the *2018 Latinx in the Mathematical Sciences Conference* at the Institute for Pure and Applied Mathematics (IPAM), Los Angeles, USA, March, 2018.

GRANT ACTIVITY AND AWARDS

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| Covariate Dependent Networks for Multi-Subject Longitudinal Studies | November, 2025 |
| <i>2025 Early Career Award Program for Assistant Professors</i> | |
| <i>Direct cost budget: \$50,000 USD. Not granted.</i> | |
| Testing Effects of Moderators on Multiple Behavior Networks | March, 2026 |
| <i>2026 UCR Research Seed Funding Small Grant Program.</i> | |
| <i>Role: Principal investigator. Direct cost budget: \$16,387 USD. Pending.</i> | |
| Detection of Common Hubs Across Multiple Gaussian Graphical Models | March, 2026 |
| <i>2026 Academic Senate Omnibus Grant</i> | |
| <i>Role: Principal investigator. Direct cost budget: \$1,000 USD. Pending.</i> | |
| Analyzing Moderator Effects on Cross-subject Brain Networks | April, 2026 |
| <i>2026 Bavarian-Californian Technology Center Grant</i> | |
| <i>Role: Co-PI. Direct cost budget: \$20,000 USD. In development.</i> | |